

Spectral Clustering

1 Introduction

The goal of this project is the complete implementation of the *spectral clustering* method for undirected graphs, using exclusively classical numerical methods for calculating eigenvalues and eigenvectors. The problem is approached through the spectral analysis of the graph's Laplacian matrix and the application of numerical algorithms such as Householder transformations and the implicit QR iteration with Wilkinson shift.

2 Mathematical Modeling of the Graph

Consider an undirected graph

$$G = (V, E), \quad |V| = n,$$

where $V = \{1, 2, \dots, n\}$ is the set of vertices, and E is the set of edges.

The graph is represented by its adjacency matrix:

$$A = (a_{ij})_{i,j=1}^n, \quad a_{ij} = \begin{cases} 1, & (i, j) \in E, \\ 0, & \text{otherwise.} \end{cases}$$

The degree matrix is defined as:

$$D = \text{diag}(d_1, d_2, \dots, d_n), \quad d_i = \sum_{j=1}^n a_{ij}.$$

The Laplacian of the graph is:

$$L = D - A.$$

3 The Spectral Clustering Problem

The Laplacian matrix L is symmetric and positive semi-definite, having real eigenvalues:

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n.$$

The eigenvectors corresponding to the smallest eigenvalues contain essential information about the graph's community structure. In particular:

- the eigenvector associated with λ_2 (the Fiedler vector) allows for separation into two clusters;
- the eigenvectors associated with λ_2 and λ_3 allow for separation into three clusters.

4 Reduction to an Eigenvalue Problem

The problem reduces to solving the equation:

$$Lx = \lambda x.$$

For numerical efficiency, the following steps are applied:

1. tridiagonalization of the matrix L ;
2. application of the implicit QR algorithm with shift;
3. extraction of the eigenvalues;
4. computation of the corresponding eigenvectors.

5 Tridiagonalization via Householder Transformations

For any symmetric matrix $L \in \mathbb{R}^{n \times n}$, there exists an orthogonal matrix Q such that:

$$Q^T L Q = T,$$

where T is a symmetric tridiagonal matrix.

This transformation is achieved through a sequence of Householder reflectors:

$$H_k = I - \beta_k u_k u_k^T.$$

Each reflector annihilates the subdiagonal elements in a column, preserving the symmetry of the matrix. Finally, we obtain:

$$T = Q^T L Q, \quad Q = H_1 H_2 \dots H_{n-2}.$$

6 Implicit QR Algorithm with Wilkinson Shift

For the tridiagonal matrix T , the implicit QR iteration is applied:

$$T - \mu I = QR,$$

$$T_{\text{new}} = RQ + \mu I.$$

The Wilkinson shift is defined by:

$$\mu = f_n - \frac{g_{n-1}^2}{\alpha + \text{sign}(\alpha) \sqrt{\alpha^2 + g_{n-1}^2}}, \quad \alpha = \frac{f_{n-1} - f_n}{2}.$$

This choice accelerates convergence and preserves the tridiagonal structure.

Convergence is reached when:

$$|g_i| \rightarrow 0,$$

and the diagonal elements converge to the eigenvalues of L .

7 Eigenvector Calculation

For an eigenvalue λ , the eigenvector is determined by solving a linear system:

$$(L - \lambda I)v_{k+1} = v_k,$$

followed by normalization:

$$v_{k+1} \leftarrow \frac{v_{k+1}}{\|v_{k+1}\|}.$$

This method converges rapidly when λ is known with sufficient precision.

8 Clustering Strategies

8.1 Two-Cluster Separation

Using the eigenvector v_2 , we define:

$$\text{cluster}(i) = \begin{cases} 0, & v_2(i) < 0, \\ 1, & v_2(i) \geq 0. \end{cases}$$

This separation corresponds to a minimal spectral cut.

8.2 Three-Cluster Separation

Using the eigenvectors v_2 and v_3 , the nodes are partitioned based on the signs of their components:

$$\text{cluster}(i) = \begin{cases} 0, & v_2(i) \geq 0 \wedge v_3(i) \leq 0, \\ 1, & v_2(i) < 0 \wedge v_3(i) \geq 0, \\ 2, & \text{otherwise.} \end{cases}$$

9 Validation

The obtained results are compared with the standard **SpectralClustering** implementation from **scikit-learn**, using the *Adjusted Rand Index (ARI)* to evaluate the clustering quality.

10 Conclusions

The project demonstrates the complete implementation of the spectral clustering method, using exclusively classical numerical algorithms to compute the spectrum of the Laplacian. The obtained results are consistent with standard methods and validate the correctness of the numerical approach.